2 marks unit 1

1. Define data warehousing.

👉 **Data warehousing is a process of collecting, storing, and managing large amounts of data from different sources for analysis and decision-making.**

2. What is OLAP?

👉 **OLAP (Online Analytical Processing) is a tool that helps to analyze data quickly in different ways like summaries, comparisons, and trends.**

3. Mention any two advantages of data warehousing.

👉 **(i) Provides fast access to data for decision-making.**

👉 **(ii) Stores historical data for analysis.**

4. List any two types of OLAP systems.

👉 **(i) ROLAP (Relational OLAP)**

👉 **(ii) MOLAP (Multidimensional OLAP)**

5. What is a star schema?

👉 **A star schema is a database design where a central fact table is connected to multiple dimension tables like a star shape.**

6. What is a snowflake schema?

👉 **A snowflake schema is a more complex database design where dimension tables are further split into sub-tables, forming a snowflake shape.**

7. Define data cube.

👉 A data cube is a multi-dimensional view of data that allows analysis by different dimensions like time, product, and location.

8. What is drill-down in OLAP?

👉 Drill-down means moving from summary data to more detailed data in OLAP.

9. Name any two phases in data warehouse design.

👉 (i) Data extraction

👉 (ii) Data modeling

10. What is metadata in a data warehouse?

👉 Metadata is “data about data” which explains the source, structure, and meaning of stored data.

11. What is ETL in data warehousing?

👉 ETL stands for Extract, Transform, Load – the process of collecting data, cleaning it, and loading it into the warehouse.

12. List two challenges in implementing a data warehouse.

👉 (i) High cost of implementation

👉 (ii) Data quality and consistency issues

13. What is a cloud data warehouse?

👉 A cloud data warehouse is a data warehouse stored and managed on cloud platforms instead of local servers.

14. Name any two cloud data warehouse providers.

👉 (i) Amazon Redshift 👉 (ii) Google BigQuery

15. Define data mining

👉 **Data mining is the process of finding patterns, trends, and useful information from large datasets**

unit-2

1.What is data preprocessing?

👉 Changing raw data into clean and proper form before mining.

2.Why is data preprocessing important in data mining?

👉 It improves data quality and helps get correct results.

3.What is data cleaning?

👉 Removing wrong, duplicate, or missing data.

4.Mention any two techniques used in data cleaning.

👉 (i) Fill missing values

👉 (ii) Remove noise.

5.Define missing data.

👉 Data that is not available in the dataset.

6.What is noise in data?

👉 Unwanted errors or extra information in data.

7.What is data integration?

👉 Combining data from many sources into one.

8.Mention any two challenges in data integration.

👉 (i) Different formats (schema mismatch)

👉 (ii) Repeated data (redundancy).

9.What is schema integration?

👉 Joining different database structures into one common structure.

10.What is data preprocessing?

👉 Preparing data (cleaning + transforming) before mining.

11.Why is data preprocessing important in data mining?

👉 To get accurate, complete, and useful data.

12.What is data cleaning?

👉 Correcting incomplete or wrong data.

13.Mention any two techniques used in data cleaning.

👉 (i) Smoothing data

👉 (ii) Removing duplicates.

14.What is data discretization?

👉 Changing continuous data into small ranges (intervals).

15.Name two methods of discretization.

👉 (i) Binning

👉 (ii) Histogram method.